What Do School Leaders Do?

Evidence from mandatory principal turnover in Spanish primary schools

Paper idea for discussion. October 2026

What does a school leader change, and how far does it reach?

Principals are an input in the production of education, and their value added is well documented. How leadership turns into teacher behavior, classroom organization, family engagement, and learning is not.

What we know

Leaders matter	Principal value added: 0.05 to 0.10 SD of achievement per SD of principal quality (Branch, Hanushek, and Rivkin, 2012; Böhlmark, Grönqvist, and Vlachos, 2016); smaller once within-school variation is cleaned (Bartanen, Husain, and Liebowitz, 2024).
Changes matter	Principal turnover lowers achievement by about 0.03 SD and raises teacher turnover (Bartanen, Grissom, and Rogers, 2019).
The channel so far	Personnel: effective principals retain strong teachers and shed weak ones (Grissom and Bartanen, 2019); principals reshape the staff through hiring and retention (Bartanen and Grissom, 2023).
Not yet seen	Under an exogenous change of leader: the same teachers' practice and wellbeing, the classrooms, and the families.

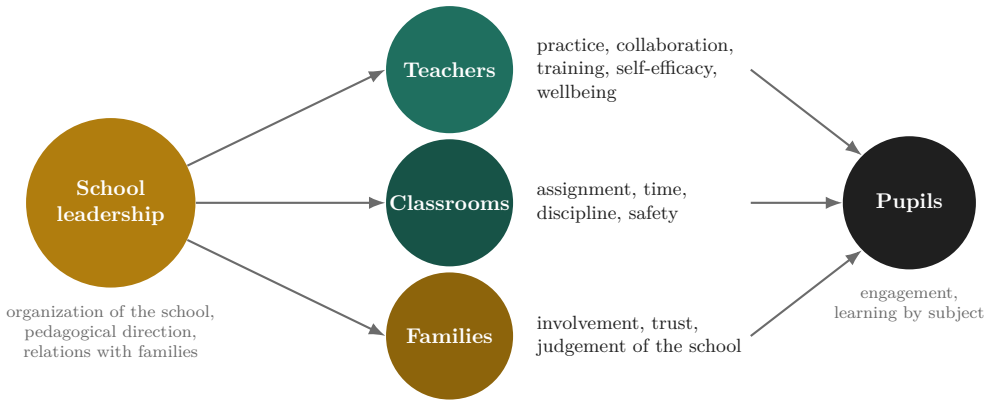
Recent work

What a leader can change, and what we observe

Teachers	Classroom practice, collaboration, training, self-efficacy, wellbeing.
Classrooms	Allocation of teachers and support hours across groups, discipline, safety.
Families	Involvement, satisfaction with teachers and the school, perceived quality.
Pupils	Engagement, perceptions of the school, achievement by subject.

Each is measured for every school and year. Teacher and family outcomes are dependent variables in their own right, and jointly the channels to achievement.

Causal map



All nodes observed for the universe of schools. Exogenous variation enters at the leadership node.

Setting

Institutions	Principals are teachers selected from the staff for four-year terms. Staffing, pay, and curriculum are set by the administration: a new principal changes leadership on a fixed staff.
Exogenous timing	Terms end by compulsory retirement at 65 and a twelve-year limit, both fixed years in advance.
Successor	Comes from the same staff; observed as a classroom teacher the year before appointment.
Data	Universe of Madrid public primary schools, 2015 to 2018: principal and teacher questionnaires, timetables, all pupils and parents by classroom, standardized tests by subject.

Identification: a principal's clock



Treated	Schools whose clock runs out in 2016, 2017, or 2018.
Comparison	Schools whose clock runs out later. Staggered event study; 2SLS for the realized change.
Benchmark	Non-rule exits (resignations, non-renewals), kept separate: the selection the literature worries about (Miller, 2013).
Heterogeneity	Successor's pre-appointment practices and wellbeing; committee score of their project; staff assessment of the outgoing principal.

The clock, in one equation

Let τ_s be the year school s's clock runs out: its principal turns 65, or completes year twelve.

$$Y_{ist} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\, t - \tau_s = k \,] + \varepsilon_{ist}$$

i	A teacher, a classroom, a pupil, or a parent in school s in year t.
$\beta_k,\ k \geq 0$	The effect k years after a rule-based change of principal, relative to the year before.
$\beta_k,\ k < 0$	Must be zero. The test that the change is not a response to the school.
Controls	Schools whose clock has not yet run out. Clustered by school; staggered-robust weights.

Idea 1: the effect of a change of leader

$Z_{st} = \mathbf{1}[t \geq \tau_s]$ is the rule; $D_{st} = \mathbf{1}[\text{new principal in } s \text{ at } t]$ is what happened.

$$D_{st} = \pi Z_{st} + \alpha_s + \lambda_t + u_{st}$$
$$Y_{ist} = \beta \widehat{D}_{st} + \alpha_s + \lambda_t + \varepsilon_{ist}$$

β	The effect of a new principal in schools where the rule forced the change.
π	One at 65. Below one at twelve years: the principal may stand again and win.
Same people	Replace α_s by α_j for the same teacher under two principals; add the grade-3 score for the same pupil in grade 6.

Idea 2: do leaders differ at all?

If $Y_{st} = \mu_s + \theta_{p(s,t)} + e_{st}$, a rule-based transition replaces θ_{old} by θ_{new}:

$$\operatorname{Var}\big(\Delta Y_{st} \mid Z_{st} = 1\big) - \operatorname{Var}\big(\Delta Y_{st} \mid Z_{st} = 0\big) = 2\,\sigma_\theta^2$$

σ_θ^2	The variance of leader effects on each outcome: how much leaders differ, without ranking any of them (Jones and Olken, 2005).
Assumption	e_{st} is unrelated to the clock. The same pre-trend test as before.
Reading	A zero mean effect with a large σ_θ^2 means leaders matter and cancel out on average.

Idea 3: what makes a good leader

$$Y_{ist} = \alpha_s + \lambda_t + \beta\, D_{st} + \delta\, D_{st} \times W_s + \varepsilon_{ist}$$

W_s	Fixed before the new principal takes office: their own practices and wellbeing as a teacher in $\tau_s - 1$; the committee's score of their project; the staff's rating of the outgoing principal in $\tau_s - 1$.
δ	How the effect of the change varies with who leaves and who arrives.
Check	W_s does not predict $\beta_{k<0}$: the type of the successor is unrelated to where the school was going.

Outcomes and expected timing

Outcome	What a new leader may change	Measured with	When
Classrooms	Assignment of teachers and support hours across groups	Schedules, class composition	Year 1
Classrooms	Discipline and climate	Pupil and teacher reports	Years 1–2
Families	Involvement, trust, judgement of the school	Parent survey	Years 1–2
Teachers	Self-efficacy, wellbeing, collaboration, training	Teacher survey	Years 1–2
Teachers	Classroom practice of the same teachers	Teacher survey, within teacher	Years 2–3
Pupils	Engagement and learning, by subject	Pupil survey, standardized tests	Last

Pre-registered indices by domain. Achievement is the final outcome; intermediate domains should respond first.

Data

In hand	Universe of Madrid public primary schools, 2015 to 2018. Principal questionnaire. Teachers: practices, wellbeing, training. Schedules. All pupils by classroom: scores by subject, perceptions. All parents by classroom: involvement, satisfaction.
To collect	Principal appointments, renewals, exits with cause (BOCM). Committee score of each project. Teacher assignment lists: chosen versus imposed posts. School plans from archived websites. Public school-level results from 2022.

Open questions

- **Mechanisms to prioritize.** Which intermediate outcomes should respond within the first year of a new principal, and which later?
- **What the data must have.** A stable teacher identifier across years; the principal's age and years in office; the survey items that rate the principal.
- **Sample of rule-based transitions** in 2015 to 2018, and access to appointment records and later years.

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Appendix: recent work on school and public-sector leaders

Title	Journal	Year
Principal effects on teacher working conditions (Rahaman and Koedel)	CALDER WP	2026
Rethinking principal effects on student outcomes	J. Public Economics	2024
Managers' productivity and recruitment in the public sector: the case of school principals	AEJ: Economic Policy	2024
Personnel management and school productivity: evidence from India	Economic Journal	2024
The lasting impacts of middle school principals	NBER WP	2024
Managers in public schools (Tsao)	Princeton JMP	2024
Effects and mechanisms of CEOs quality in public education	Economic Journal	2023
School principal race, teacher racial diversity, and student achievement	J. Human Resources	2023
Managers and productivity in the public sector	Econometrica	2022

Tsao: event studies of principal moves and retirements, Kentucky and Texas; effects on scores, absenteeism, teacher retention. Rahaman and Koedel: variance of principal effects on 20 teacher-reported working conditions, Illinois.

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